Advanced Topics in Deep Learning

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Specialized Learning

- Structure
  - Custom model architectures designed for each task
  - Trained on small, annotated datasets
  - Strong inductive bias, lightweight, easy deployment

- Limitations
  - Requires expert design per task
  - Heavy reliance on annotated data
  - Limited reuse of knowledge across tasks
  - Expensive to maintain many models

Transfer Learning

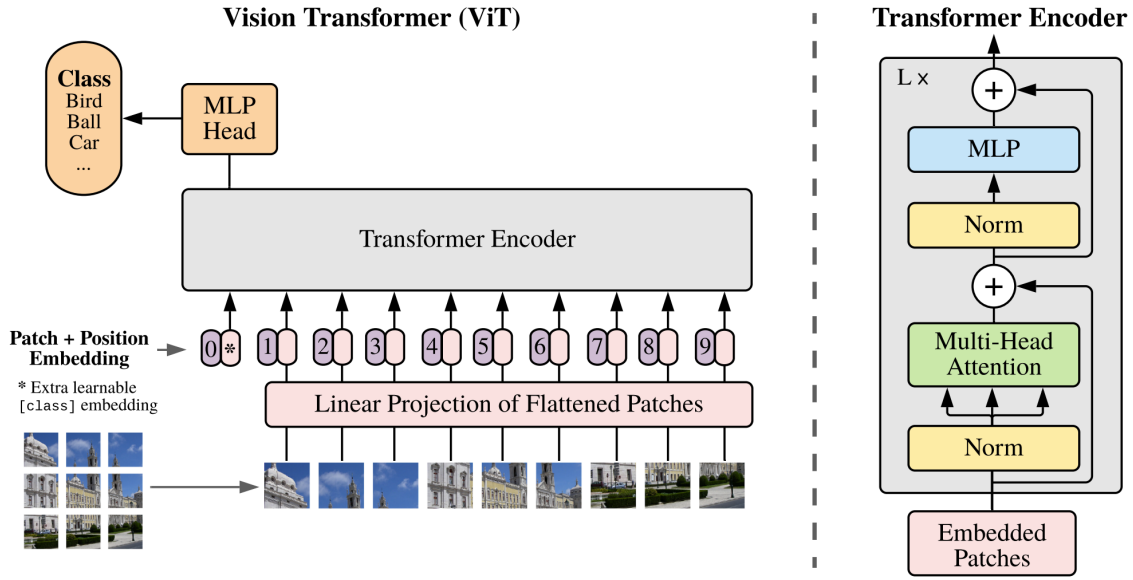
- Structure
  - Pretrain on large-scale data (e.g., ImageNet, BERT pretraining)
  - Reuse learned representations for downstream tasks
  - Add task-specific modules, fine-tune with smaller datasets
- Benefits
  - Better generalization than specialized models
  - Reduces labeled data requirements
  - Accelerates training and improves robustness
- Limitations
  - Often requires task/domain-specific adaptation
  - Performance may degrade on out-of-distribution (OOD) data
  - Pretraining and adaptation are task/domain-specific (vision, language, etc.)

Foundation Models

- Structure
  - Pretrained on large-scale data using self-supervised or weakly supervised objectives
  - Unified, general-purpose backbones across domains
  - Adaptable via prompting, probing, adapters, or few-shot learning
- Advantages
  - Scalability: knowledge transfers across many tasks/domains
  - Efficiency: reduce need for task-specific design and annotation
  - Versatility: same model may support multiple modalities (text, image, speech, code)
- Challenges
  - High compute and energy cost
  - Risks: bias, misuse, lack of interpretability
  - Still requires domain adaptation for sensitive tasks (e.g., medicine)

ViT

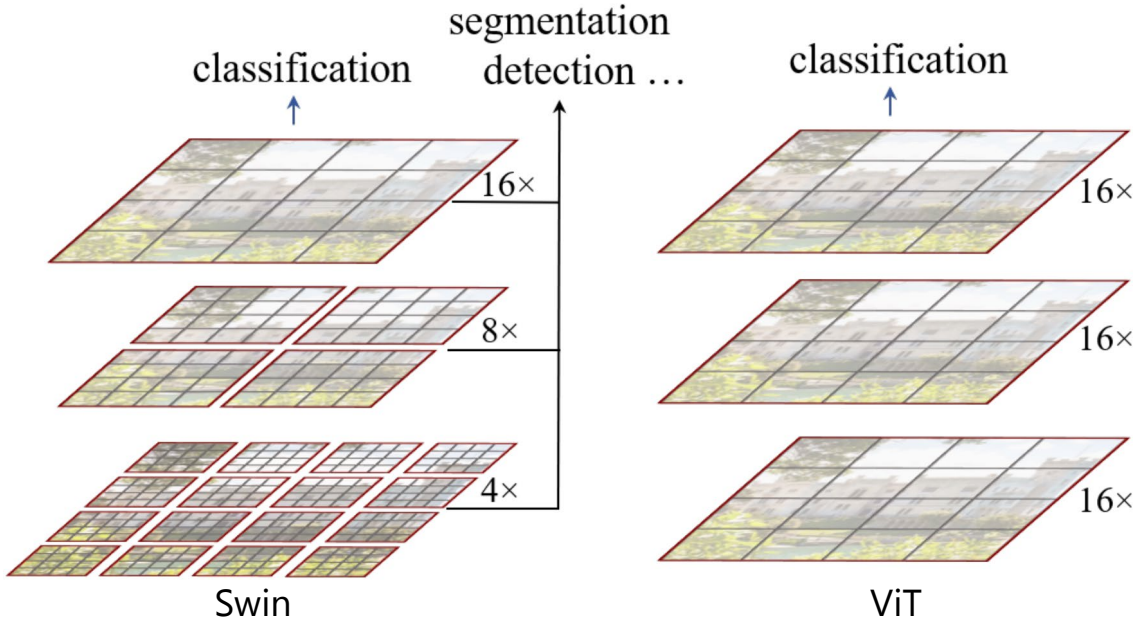
- Vision Transformers for image recognition
  - Images split into fixed-size patches, flattened, and positionally embedded into tokens
  - Transformer encoder learns spatial structure, which lacks key CNN inductive biases such as locality and translation equivariance
  - Efficiency at scale (superior performance with large-scale training) not crafted biases

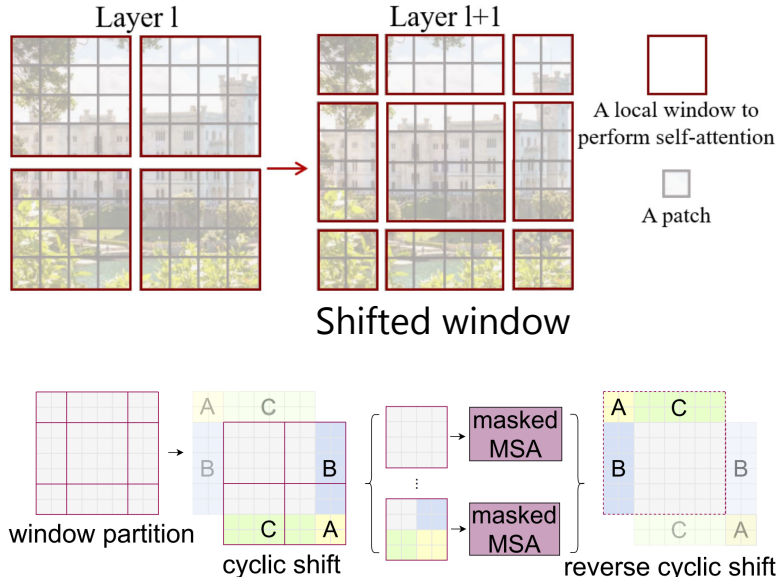


https://arxiv.org/pdf/2010.11929v2
https://github.com/google-research/vision_transformer

Swin Transformer

- Hierarchical ViTs using shifted windows
  - Multiscale feature maps by merging neighboring image patches in deeper layers
  - Self-attention in non-overlapping windows & shifting window partition across layers
  - Local attention supports high-resolution images with linear compute scaling

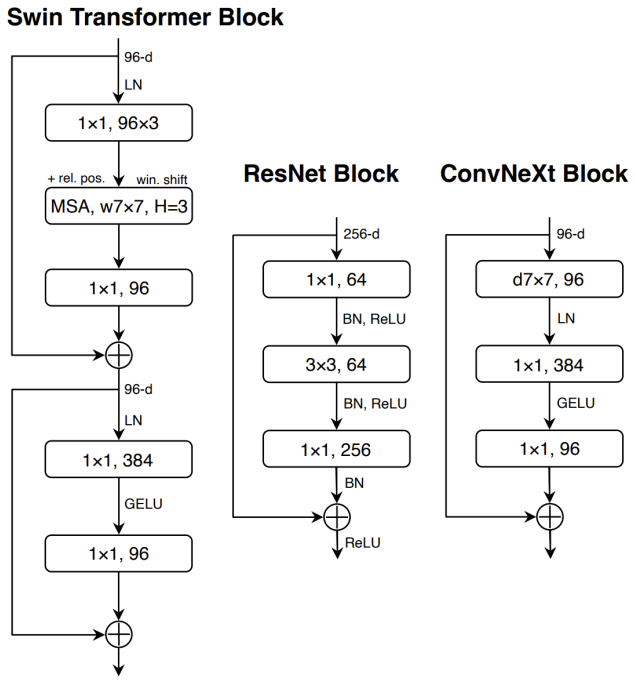


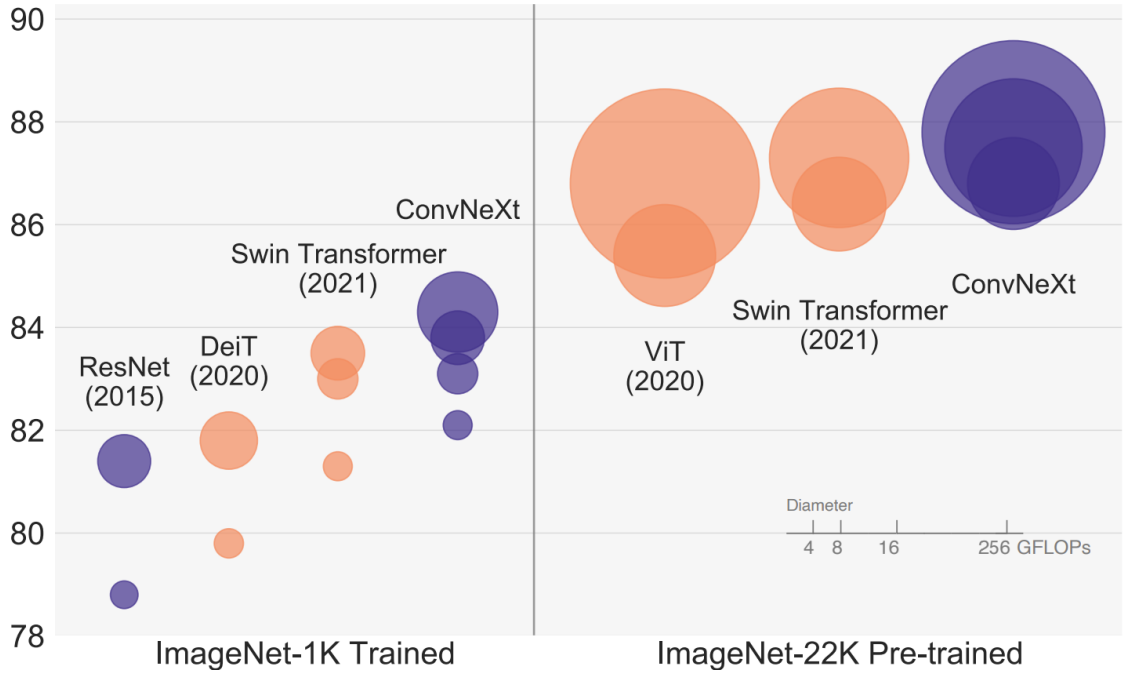


https://arxiv.org/pdf/2103.14030v2
https://github.com/microsoft/Swin-Transformer

ConvNeXt

- Convolutional network design explorations inspired by ViTs
  - ResNet design revisions: inverted bottlenecks, larger kernel sizes, streamlined depth
  - A general-purpose backbone: high performance with low architectural risk

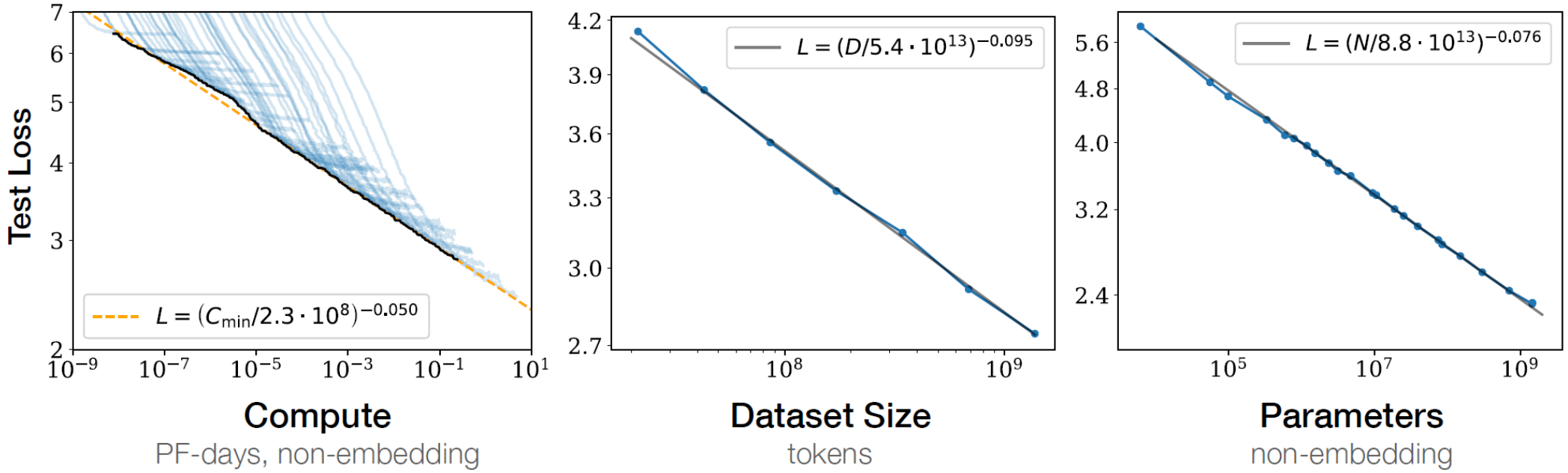




https://arxiv.org/pdf/2201.03545v2
https://github.com/facebookresearch/ConvNeXt

Kaplan Scaling Laws

- Designing optimally efficient models under resource constraints
  - Training loss follows predictable power-law patterns, independent of model size
  - Test loss follows a power-law relation with model size, dataset size, and compute
    - For optimal performance, all three factors must be scaled up in tandem



https://arxiv.org/pdf/2001.08361
https://github.com/shehper/scaling_laws

Kaplan Scaling Laws

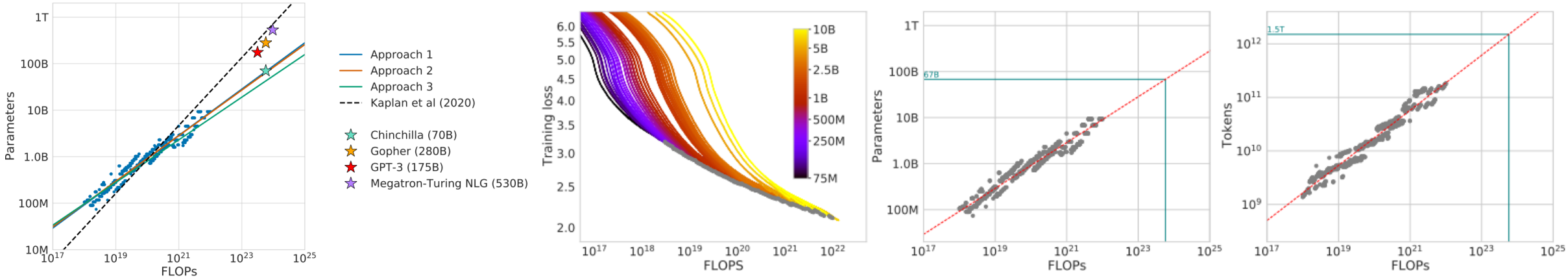
- Designing optimally efficient models under resource constraints
  - Performance depends on scale, architectural variations have minimal impact
  - Bigger models reach better performance more quickly and with fewer examples
  - Train very large models on moderate data size, stop before full convergence

Parameters	Data	Compute	Batch Size	Equation
N	∞	∞	Fixed	$L(N) = (N_c/N)^{\alpha_N}$
∞	D	Early Stop	Fixed	$L(D) = (D_c/D)^{\alpha_D}$
Optimal	∞	C	Fixed	$L(C) = (C_c/C)^{\alpha_C}$ (naive)
N_{opt}	D_{opt}	$C_{\min}$	$B \ll B_{\text{crit}}$	$L(C_{\min}) = \left(C_c^{\min}/C_{\min}\right)^{\alpha_C^{\min}}$
N	D	Early Stop	Fixed	$L(N, D) = \left[\left(\frac{N_c}{N}\right)^{\frac{\alpha_N}{\alpha_D}} + \frac{D_c}{D}\right]^{\alpha_D}$
N	∞	S steps	B	$L(N, S) = \left(\frac{N_c}{N}\right)^{\alpha_N} + \left(\frac{S_c}{S_{\min}(S,B)}\right)^{\alpha_S}$

https://arxiv.org/pdf/2001.08361
https://github.com/shehper/scaling_laws

Chinchilla Scaling Laws

- Designing optimally efficient models under resource constraints
  - Traditional scaling laws
    - increasing model size → improved performance
    - training data is coupled with model size and scales proportionally
  - Current scaling laws
    - performance improvements can be achieved by balancing model size, training data, and compute
    - LLMs could be substantially smaller and trained much longer than expected



https://arxiv.org/pdf/2203.15556
https://github.com/kyo-takano/chinchilla

SSL

- Self-supervised pretraining
  - Learning context-aware representations without labeled data for downstream tasks
  - Training objectives are derived from the data itself
    - e.g., masked patch prediction, consistency between augmented versions, representation matching

- Benefits
  - Bypasses the need for manual labels
  - Matches or surpasses supervised methods on benchmarks like ImageNet
  - Scales across modalities (text, vision, video, audio, ⋯)

https://arxiv.org/pdf/2304.12210v2

SSL Methods

- Deep metric learning (contrastive)
  - Contrast views of the same input (positives) and push apart others (negatives)
  - Good for simplicity, works well with strong augmentations and negatives
  - SimCLR, MoCo, NNCLR, MeanSHIFT, SCL

- Representation matching (non-contrastive)
  - Match representations by predicting one view from another (no negatives)
  - Avoids needing negatives, simpler pipelines with strong performance
  - DINO, BYOL, SimSiam

https://arxiv.org/pdf/2304.12210v2

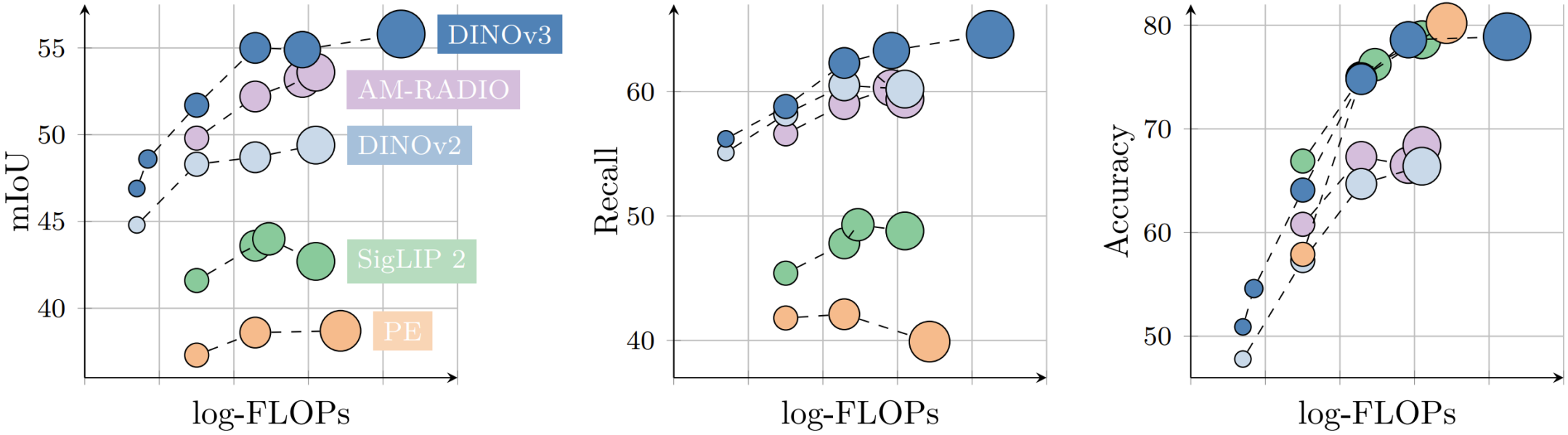
SSL Methods

- Canonical correlation analysis (CCA)
  - Encourage alignment between views while preserving diversity (avoiding collapse)
  - Mathematically robust, balances alignment/invariance and variance preservation
  - VICReg, Barlow Twins, W-MSE
- Masked image modeling (MIM)
  - Mask part of the input and predict pixels, discrete tokens, or latent representations
  - Excels with transformers, strong in dense tasks and data efficiency
  - MAE, BEiT, IBOT
- Joint-embedding prediction (JEPA)
  - Predict representations of missing/masked content blocks rather than raw pixels
  - I-JEPA, V-JEPA

https://arxiv.org/pdf/2304.12210v2

DINOv3

- DINOv3: Scalable self-supervised vision foundation model by Meta AI
  - Trained on 1.7B images, 7B parameters, zero labels
  - Gram anchoring regularization, post-training/distillation for universal visual features

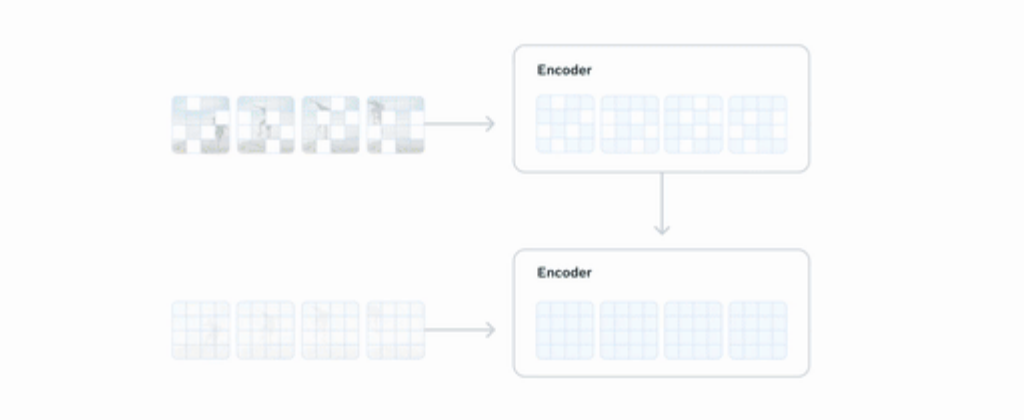


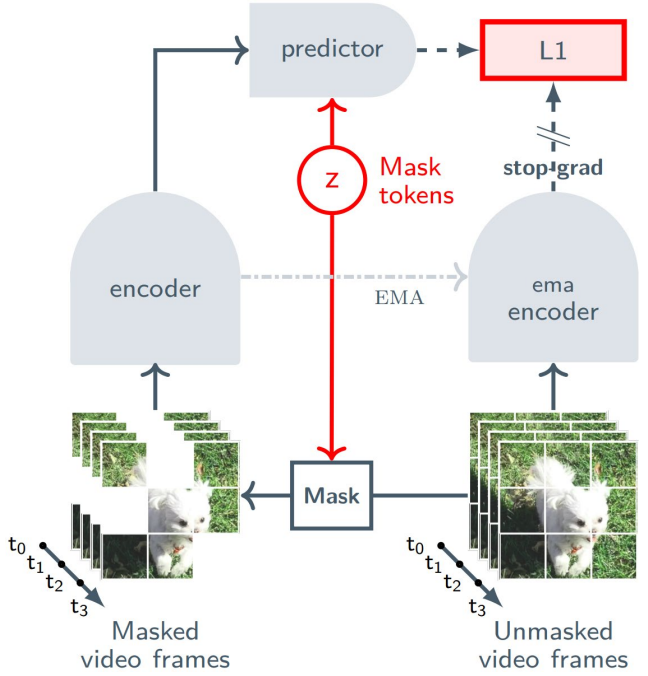
(a) Semantic segmentation (ADE20k) **(b)** 3D keypoint matching (NAVI) **(c)** OOD classif. (ObjectNet)

https://arxiv.org/pdf/2508.10104
https://github.com/facebookresearch/dinov3

V-JEPA 2.1

- Video-based Joint-Embedding Predictive Architecture
  - Learning dense, high-quality visual representations for both images and videos while retaining strong global scene understanding

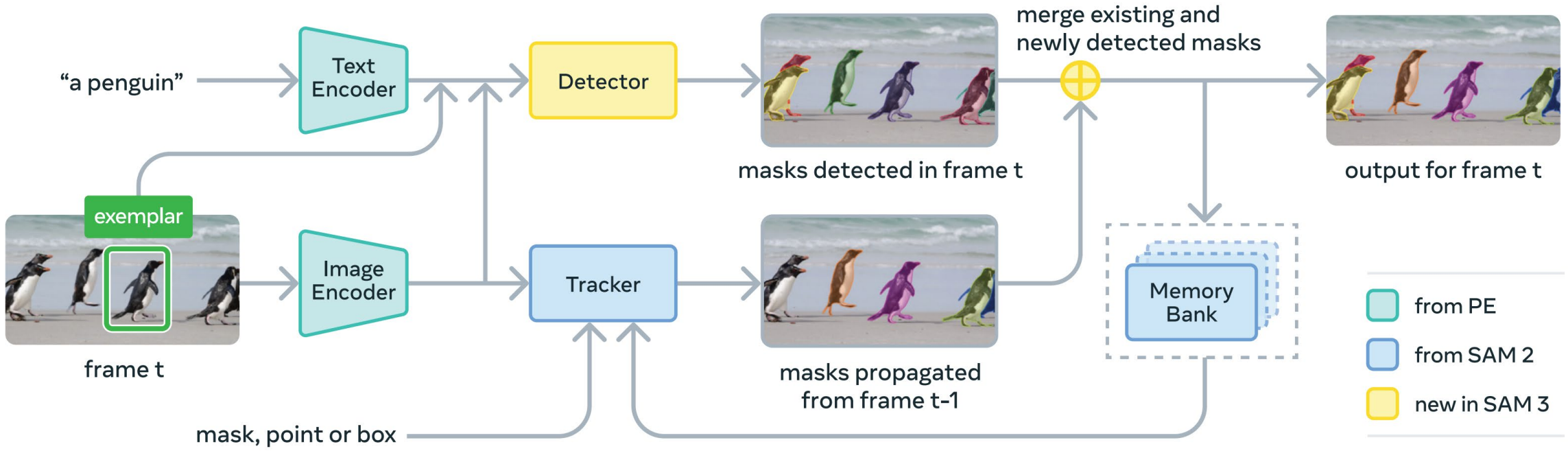




https://arxiv.org/pdf/2603.14482v3
https://github.com/facebookresearch/vjepa2

SAM 3

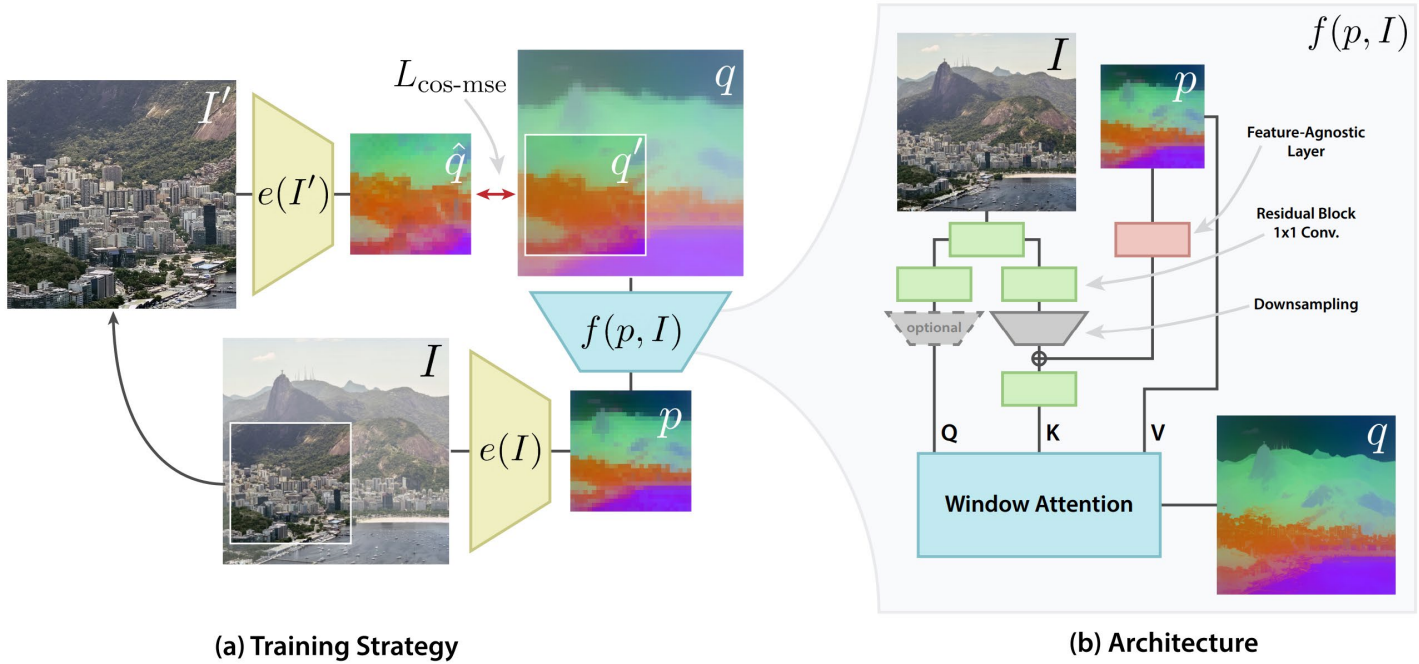
- Segment Anything with Concepts
  - A unified model detecting, segmenting, and tracking objects in images or videos based on concept prompts



https://arxiv.org/pdf/2511.16719v2
https://github.com/facebookresearch/sam3

AnyUp

- A universal model that can upsample features from any layer & resolution
  - A window attention-based upsampling architecture trained using an image part-based loss, retaining the input feature space through consistency regularization



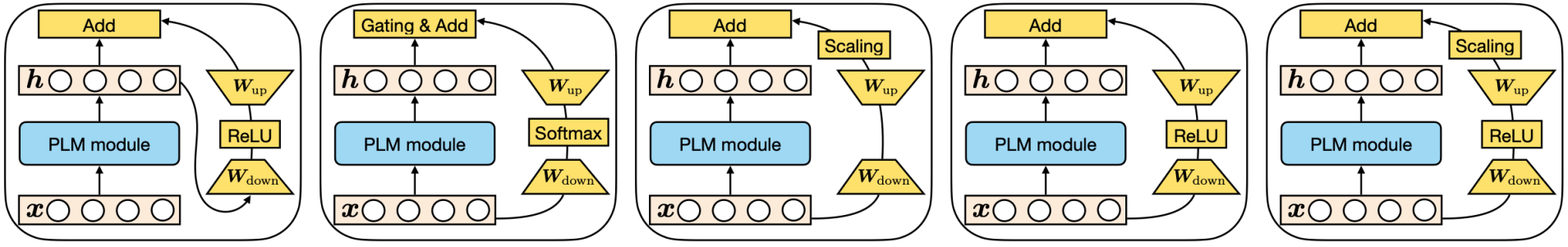
(a) Training Strategy

(b) Architecture

https://arxiv.org/pdf/2510.12764v2
https://github.com/wimmerth/anyup

Parameter-Efficient Transfer Learning

- A unified view for parameter-efficient tuning

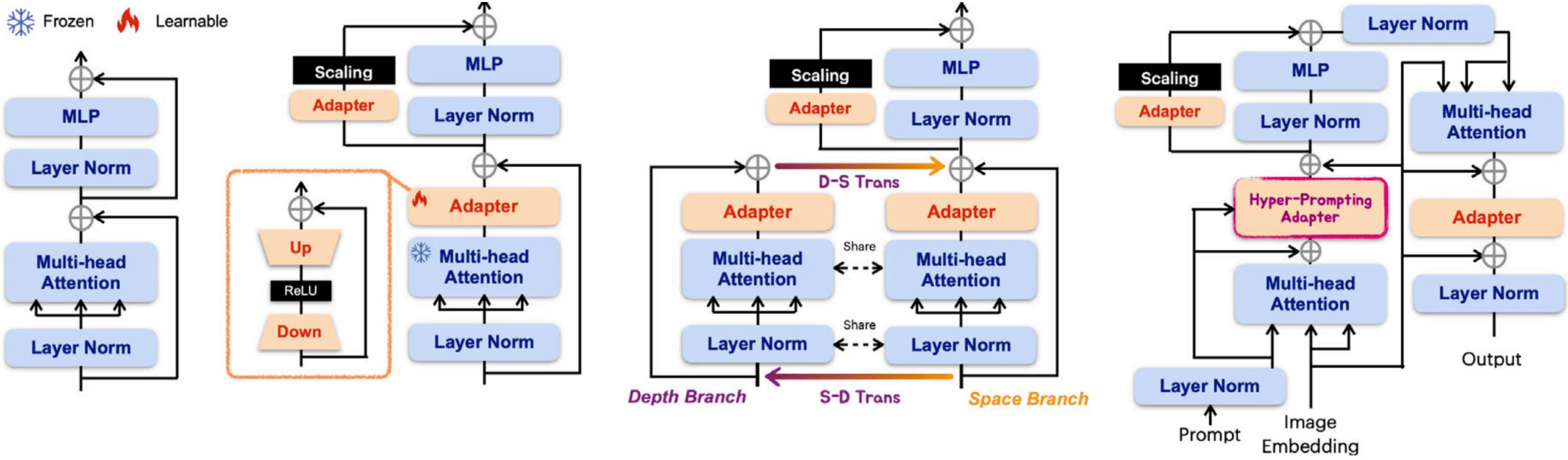


(a) Adapter (b) Prefix Tuning (c) LoRA (d) Parallel Adapter (e) Scaled PA

https://arxiv.org/pdf/2110.04366v3
https://github.com/jxhe/unify-parameter-efficient-tuning
https://github.com/huggingface/peft

Medical SAM Adapter (Med-SA)

- Adapting SAM for medical image segmentation



a. ViT Block in SAM b. 2D Medical Image Adaption c. 3D Medical Image Adaption d. Decoder Adaption with Prompt

https://arxiv.org/pdf/2304.12620v7
https://github.com/ImprintLab/Medical-SAM-Adapter

Thank you!

Any questions?